

Practice Problems-8

Young's experiment is performed with orange light from a krypton arc. If the fringes are measured with a micrometer eyepiece at a distance 100 cm from the double slit, it is found that 25 of them occupy a distance of 12.87 mm between centers. Find the distance between the centers of the two slits.

Ans. 1.1297 mm

When a thin film of transparent plastic is placed over one of the slits in Young's double-slit experiment, the central bright fringe, of the white-light fringe system, is displaced by 4.50 fringes. The refractive index of the material is 1.480, and the effective wavelength of the light is 5500 Å. (a) By how much does the film increase the optical path? (b) What is the thickness of the film? (c) What would probably be observed if a piece of the material 1.0 mm thick were used? (d) Why?

Lloyd's-mirror experiment is readily demonstrated with microwaves, using as a reflector a sheet of metal lying flat on the table. If the source has a frequency of 12,000 MHz and is located 10.0 cm above the sheet-metal surface, find the height above the surface of the first two maxima 3.0 m from the source.

Ans. (a) 18.750 cm, (b) 56.25 cm

Note: A phase change of π occurs upon reflection; see Sec. 13.6.

Find the angular radius of the tenth bright fringe in a Michelson interferometer when the central-path difference ($2d$) is (a) 1.50 mm and (b) 1.5 cm. Assume the orange light of a krypton arc is used and that the interferometer is adjusted in each case so that the first bright fringe forms a maximum at the center of the pattern.

Ans. (a) 4.885°, (b) 1.542°